

LUKAS UZOLAS

Zürich, Switzerland

Software Development & Research

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Summary Extensive knowledge of 2D/3D machine learning and deep learning workflows in Python. Specialization in image synthesis and 3D object dynamics. Experience working in international and dynamic environments. Enjoy working on open-ended problems that require creative thinking and collaboration.

EDUCATION

Ph.D. Computer Science @ CGV Delft University of Technology	May 2022 – May 2026 Delft, Netherlands
Topics: Neural Rendering, Dynamic 3D Reconstruction, 3D Animation, Generative Modelling	
M.Sc. Image Processing and Computer Vision University of Bordeaux	Sep. 2019 – Sep. 2021 Bordeaux, France
Autonomous University of Madrid	Madrid, Spain
Pázmány Péter Catholic University	Budapest, Hungary
Awards: Graduated best of class, Erasmus Mundus Master Scholarship recipient	
B.Sc. Human-Computer-Interaction University of Hamburg	Oct. 2015 – Jun. 2019 Hamburg, Germany
Awards: Graduated best of class	

EXPERIENCE

Research Intern Google	May. 2025 – Aug. 2025 Zurich, Switzerland
- Improved the controllability of pre-trained diffusion models with synthetic 3D data - Implemented large-scale pre-processing, training, and evaluation pipelines for 2D diffusion models in JAX and Tensorflow - Investigated and proposed research directions to stakeholders independently	
Ph.D. Candidate Delft University of Technology	May 2022 – ongoing Delft, Netherlands
- Formulated research directions in 3D Computer Vision and Graphics, and consistently delivered projects on time, resulting in multiple publications at high-impact venues - Guided students during their thesis projects as a co-supervisor during weekly meetings, helping with research questions and methodology, leading to 5 successful B.Sc. and M.Sc. graduations - Assisted students with advanced imaging concepts as a teaching assistant in <i>Applied Image Processing</i> and <i>3D Visual Computing</i> in Python and C++, and created assignments and grading pipelines allowing for efficient evaluation of 100+ students at a time	
Research Intern (remote, due to Covid pandemic) Harvard University	Feb. 2021 – Jun. 2021 Cambridge, MA, USA
- Initiated a collaboration between Visual Computing Group at Harvard and the University of Bordeaux, resulting in my Master's thesis on Domain Adaptation for medical image segmentation - Developed and evaluated a method for making normalization layers in deep neural networks more invariant to the domain gap present between training and inference datasets	
Research Intern MEDIC	Jul. 2020 – Sep. 2020 Madrid, Spain
Explored and implemented algorithms for extracting features from high frame rate videos of eyes, for improving ocular refractive index calculations for simpler diagnostics of vision impairments	

Working Student Fullstack Development (*part-time*)

Oct. 2017 – Feb. 2019

Senacor, Daimler AG, BOOM GmbH*Hamburg & Stuttgart, Germany*

- Conceptualized and implemented web applications, developing backend logic in Node.js & C#, HTTPS APIs with Express, and creating user interfaces with Javascript & Vue.js
- Demonstrated the technical feasibility of solutions to stakeholders by proposing and realizing PoCs such as a intelligent car configurator, WhatsApp chatbots, and marketing apps for clients such as Nivea and Germany Travel

Student Supervisor in Software Development I & II (*part-time*)

Oct. 2016 – July. 2017

University of Hamburg*Hamburg, Germany*

- Taught and examined software development and programming concepts in Java

SELECTED PUBLICATIONS

Rotation-Equivariant Neural Jacobian Fields for Generalizable 3D Mesh Deformations, SIGGRAPH Asia 2026Lukas Uzolas, Ruben Wiersma, Olga Sorkine-Hornung, Petr Kellnhofer**Topics:** Geometry Processing, 3D Animation**Surface-Aware Distilled 3D Semantic Features**, SIGGRAPH Asia 2025 [[paper](#), [site](#)]Lukas Uzolas, Elmar Eisemann, Petr Kellnhofer**Topics:** Self-Supervised Learning, Geometry Processing, Foundation Models**MotionDreamer: Exploring Semantic Video Diffusion features for Zero-Shot 3D Mesh Animation**, 3DV 2025 [[paper](#), [site](#)]Lukas Uzolas, Elmar Eisemann, Petr Kellnhofer**Topics:** Generative Models, Diffusion Models, Differentiable Rendering, User Studies, PyTorch**Template-free Articulated Neural Point Clouds for Reposable View Synthesis**, NeurIPS 2023 [[paper](#), [site](#)]Lukas Uzolas, Elmar Eisemann, Petr Kellnhofer**Topics:** Dynamic 3D Reconstruction, Neural Rendering, NeRF, Kinematics, PyTorch**Deep Anomaly Generation: An Image Translation Approach of Synthesizing Abnormal Banded Chromosome Images**, IEEE Access 2022 [[paper](#)]Lukas Uzolas*, Javier Rico*, Pierrick Coupé, Juan C. SanMiguel, and György Cserey**Topics:** Generative Models, Generative Adversarial Networks (GANs), Synthetic Data, PyTorch

SKILLS

Domain Knowledge: Machine Learning, Deep Learning, Computer Vision, Computer Graphics, Image Processing, Neural Rendering, 3D Animation, Motion Fields, 3D Representations, 3D Reconstruction, Generative Models, Differentiable Rendering**Technologies:** Neural Radiance Fields (NeRFs), Gaussian Splatting, Scene Representations, Diffusion Models, Variational Autoencoders (VAEs), Autoregressive Models, Generative Adversarial Networks (GANs), AWS, Databases (NoSQL, MySQL), Linux**Languages:** Python, C++, JavaScript, NodeJS, Java, Matlab**Frameworks:** PyTorch, OpenCV, Scikit, Numpy, Tensorflow, JAX, Pandas

LANGUAGES

English (*Fluent*), German (*Native*), Lithuanian (*Intermediate*), Mandarin (*Beginner*)

INTERESTS

Besides enjoying running and TTRPGs, I am currently learning Mandarin, where I sometimes like to build small applications to assist my learning journey [[example](#)].